

Emotion AI Lab - Technical Documentation

Lead Engineer: Masab Rehman | Data Scientist / ML Engineer

Date: August 29, 2026

[LinkedIn Profile](#)
[GitHub Repository](#)
[Personal Portfolio](#)

1. Executive Summary

The Emotion AI Lab is an advanced Audio Signal Analysis platform engineered to classify human emotions from raw speech audio. The project tackles the complex domain of affective computing, turning variable-length audio waves into structured, actionable psychological insights.

The platform includes a specialized Inference Studio that allows users to upload audio, view extracted features, and receive instant emotion predictions.

2. Dataset & Signal Preprocessing

The system was trained on industry-standard emotional speech datasets, including EmoDB (Berlin Database of Emotional Speech) and TESS (Toronto emotional speech set).

- Noise Reduction: Applied spectral gating to remove background ambient noise from raw WAV/MP3 files.
- Normalization: Audio signals were normalized to a standard LUFS target to ensure consistent volume levels across different speakers.
- Trimming: Automatically detected and trimmed leading/trailing silence using zero-crossing rate heuristics.

3. Feature Extraction (Librosa)

The core intelligence relies on robust digital signal processing to extract psychoacoustic features.

- Mel-Frequency Cepstral Coefficients (MFCCs): Captured the timbral envelope of the voice, serving as the primary feature matrix.
- Chroma Features: Extracted pitch class profiles to analyze variations in speech tone and inflection.
- Spectral Contrast & Zero-Crossing Rate: Used to differentiate between high-energy emotions (anger/happiness) and low-energy emotions (sadness/boredom).

4. Model Architecture

The initial baseline utilized a Random Forest classifier trained on aggregated statistical features (mean,

Emotion AI Lab - Technical Documentation

Lead Engineer: Masab Rehman | Data Scientist / ML Engineer

Date: August 29, 2026

variance) of the MFCCs.

- The pipeline is designed to easily swap in deep learning components (e.g., 1D CNNs or LSTMs) for raw waveform processing.
- Multi-class classification handles emotions such as Neutral, Happy, Sad, Angry, Fearful, Disgust, and Surprised.
- Addressed class imbalance using SMOTE and weighted loss functions.

5. Full-Stack Integration

The web application features a 'Signal Processor' and 'Inference Studio' built on Flask.

- Frontend UI: A sleek, professional lab environment utilizing HTML5, TailwindCSS, and JavaScript.
- Interactive Elements: Users can upload audio files which are instantly processed by the backend, returning the predicted emotion and confidence intervals.
- Real-time Feature Visualization metrics natively embedded in the UI.